

Shivam Kumar

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EDUCATION

Vellore Institute of Technology

2023 – 2027

B.Tech in Computer Science & Engineering (Cloud Computing & Automation) | CGPA: 7.23/10 *Bhopal, India*

EXPERIENCE

FOSSEE, IIT Bombay 

4 Months

Applications & Research Intern (Internal Tools & Automation)

Remote

- Developed custom Python scripts and automated data pipelines to process 14,400+ records, successfully identifying and resolving a critical platform bug causing 92.9% of system errors.
- Collaborated with researchers to translate requirements into technical solutions, building evaluation workflows for 600+ AI responses while maintaining clean, modular, and self-sustaining code.
- Configured and deployed benchmarking scripts across 6 third-party LLM systems, optimizing API execution and generating data-driven reports for system scaling.

APPLICATION ENGINEERING PROJECTS

CertiFlow — Bulk Certificate Generator & Verification API 

Tools: FastAPI, Python, SQLAlchemy, ReportLab, SQLite, Docker, Pytest, GitHub Actions CI

- Architected an asynchronous bulk PDF generation REST API with FastAPI BackgroundTasks, providing non-blocking execution and real-time job progress polling.
- Implemented isolated fault tolerance ensuring individual malformed recipient records fail gracefully without interrupting the remaining batch.
- Developed an in-memory PDF generator using ReportLab with embedded dynamic QR verification codes linking to a public credential verification page.
- Wrote 17 automated Pytest integration tests covering validation, concurrency, and ZIP bundling; deployed on Render with GitHub Actions CI.

Autonomous RAG System & MCP Triage Agent 

Tools: Python, Model Context Protocol (MCP v2), Gemini API, ChromaDB, Gemma-3-27B, Streamlit

- Architected a production-ready Model Context Protocol (MCP v2) Server exposing semantic retrieval, vector search, and autonomous triage tools over standardized JSON-RPC stdio and public Streamable HTTP transports.
- Engineered an end-to-end RAG pipeline over 850 vector entries using 3072-dim embeddings, cosine similarity search, and Gemma-3-27B for fast, cited, and hallucination-resistant generation.
- Built an autonomous triage engine with few-shot context grounding, strict Pydantic/JSON schema validation, and deterministic multi-queue routing rules; implemented in-memory sliding-window rate limiting and LRU caching for zero-cost operation.

Priority Inbox (Internal Tooling / Chrome Extension)

Tools: JavaScript, RESTful Web Services, Gmail API

- Engineered a custom Chrome extension utilizing JavaScript and the Gmail API to parse, filter, and prioritize incoming emails based on user-defined parameters.
- Replaced manual inbox sorting with an automated REST-driven background service, demonstrating strong front-end to third-party API integration skills.

TECHNICAL SKILLS

Programming Languages: Java, Python, C++, JavaScript, SQL, HTML, CSS

Application Engineering: API Development, RESTful Web Services, Custom Front-Ends, Back-End Services

Core CS & Algorithms: Data Structures, Algorithms, Object-Oriented Design, Systems Software, DBMS

Tools & Frameworks: Git, FastAPI, Streamlit, Vector Databases, AWS, Cloud Platforms, Model Context Protocol (MCP), Docker

ACHIEVEMENTS & CERTIFICATIONS

- **Algorithms & Problem Solving:** Secured Global Rank **67** (Starters 190) and **95** (Starters 204) on **CodeChef** ; active on **LeetCode** , showcasing strong expertise in algorithms.
- **Global Rankings:** Consistently ranked among the top global participants with Ranks **188** and **253** in CodeChef Starters 195 and 196 respectively.
- **Certifications:** **Machine Learning** (UoM) , **Google IT Support Professional** , **Internet of Things** (NPTEL) .